Power Outage Forecasting and Backup Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Spring 2026

1 Introduction

Power-grid *resilience* — the capability of withstanding, adapting to, and recovering from large-scale disruptions — has become a top priority as climate-driven extreme weather grows more frequent. The April 2025 Pittsburgh windstorm left 140k customers without power for 24+ hours despite 400 line workers in the field; Hurricanes Beryl and Milton each downed 3M+ Florida customers in 2024. **Proactive (ex-ante) actions** — pre-positioning repair crews and mobile generators before forecasted events — are the operational lever utilities can pull, but they require accurate short-horizon outage forecasts. We address two integrated tasks: (i) 24-hour and 48-hour hourly outage forecasts for the 83 Michigan counties, and (ii) allocation of five 1,000-household mobile generators to the counties where they will absorb the most outage on the test horizon.

We are given an hourly county-level dataset with 109 atmospheric/hydrological features over a 90-day window (**2023-04-01** – **2023-06-30**). Two design constraints shape the entire pipeline. **No future weather is available at test time**: only outage history and weather observed up to the forecast origin can enter the model — mirroring real deployment, where utilities likewise lack ground-truth future weather at decision time. **The training window ends mid-storm**: the largest event of the year (peak 86,313 outages on 2023-06-29) starts 2023-06-23 and runs through the end of training, so the test horizon begins inside an unfolding storm. These constraints directly motivate our validation design (§3.1) and feature engineering (§3.2).

2 Data Analysis

The data is shape T=2,161 $\times$ L=83 $\times$ F=109, plus an outage matrix $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{T \times L}$ and a per-county `tracked` household count.

Sparse, heavy-tailed target. 70.5% of (c, t) cells are zero, and the non-zero distribution is heavy-tailed (Fig. 2a). State-wide hourly outage ranges 105–86,313, with the 99.5th percentile 10$\times$ the median — a zero-inflated, log-normal-tailed regression target.

Storm-event structure. Using a state-wide threshold of 2,000 simultaneous outages with a 6-hour minimum duration (12-hour merge gap), we detect **26 storm events** in the 90 days (Fig. 1). The largest event (id 25, 2023-06-23 – 06-30) peaks at 86,313, more than 3$\times$ the second-largest (27,643). *Training ends inside this event.*

Strong autocorrelation. The state-wide series has a clear diurnal echo at lag 24h and a weaker weekly echo at 168h (Fig. 2b). Lagged outage features should dominate; weather is informative but secondary, as much of its predictive value is already encoded in lagged outage.

County heterogeneity. Per-county `tracked` ranges ~5,000 to 920,000. Wayne County (FIPS 26163, Detroit) tracks 922k households but its peak hourly outage of 15,143 is only third largest; smaller counties like Saginaw (97k tracked, 8.7k peak) take disproportionately large per-capita hits in storm events (Fig. 2c). This drives our policy recommendation (§5).

Top weather correlates. The 15 features most correlated with state-wide outage are all moisture-and-wind-related (`cape`, low-level u/v winds, precipitable water, `vstm` storm-motion magnitude). We retain all 91 non-zero-variance features; tree models implicitly select.

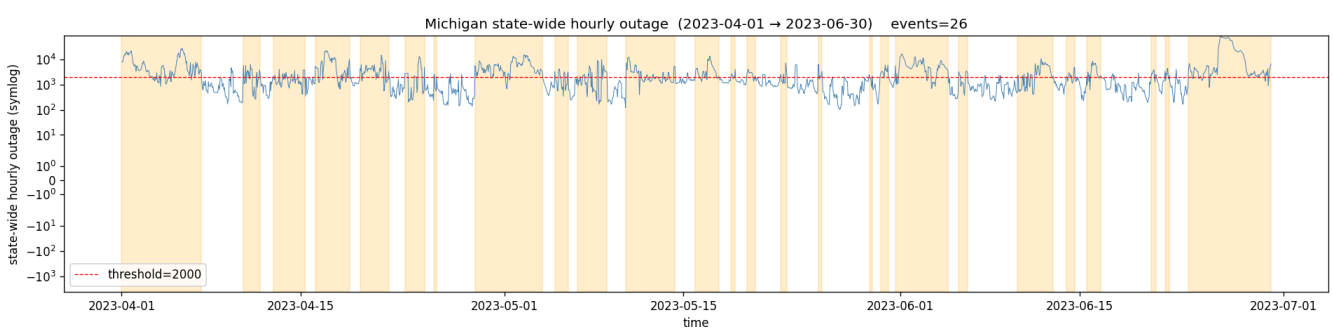
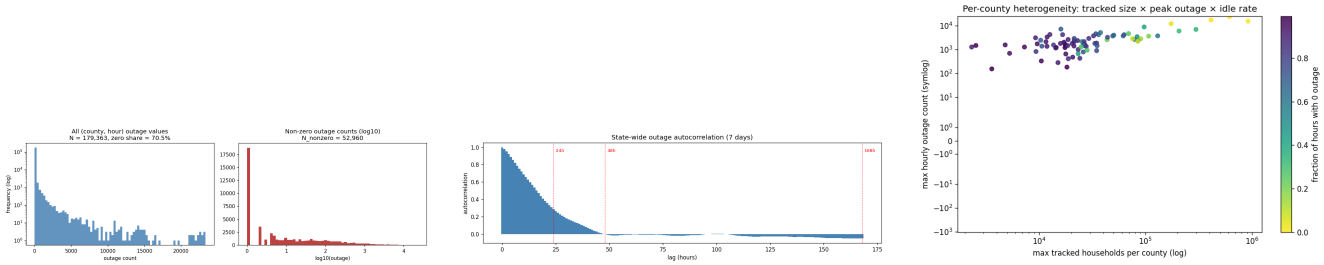


Figure 1: State-wide hourly outage on a symmetric-log scale. Orange = 26 detected storm events; red dashed = 2,000-outage threshold. The disproportionately large terminal event ends exactly at the train/test boundary.



(a) Outage value distributions. (b) Autocorrelation (7 days). (c) County heterogeneity.

Figure 2: Outage is zero-inflated and heavy-tailed (a); strongly autocorrelated at lag 24h with weaker echo at 168h (b); county sizes and storm impact vary by 2 orders of magnitude (c).

3 Methods

3.1 Validation: Rolling-Origin CV that Mirrors the Test Setup

A naive last-48h holdout falls inside the year's largest storm, so any single-window val-RMSE is a sample of one. We replace it with **rolling-origin cross-validation** (Fig. 3) of 6 folds spaced 72 hours apart. Each fold's val window is the forecast horizon (24/48 hours) immediately following its fit window — exactly the production setup. Fit windows *expand*: each fold uses all data from 2023-04-01 to its origin, so fold 0's fit set equals the final retrain on full training data, making its RMSE the most direct estimate of test performance [4]. No weather inside any val window is visible to its model. Folds span 2023-06-13 – 06-29; **exactly one fold (fold 1) lands in the big-storm peak**. We report *both* aggregate-mean and calm-period-mean RMSE: the former captures cross-period performance including OOD storms, the latter captures typical-day quality without one extreme event dominating.

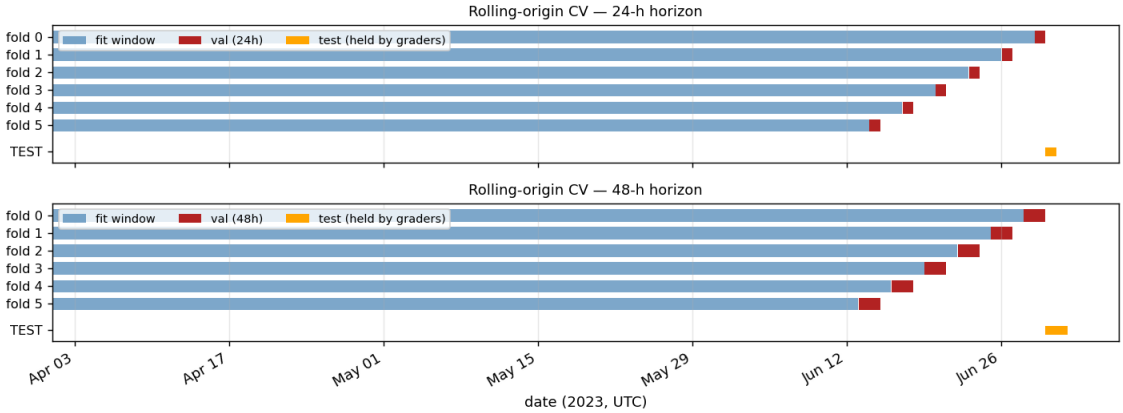


Figure 3: Rolling-origin CV layout. Blue = expanding-window fit, red = held-out validation, orange = the actual test held by the graders.

3.2 Feature Engineering

For tree-based models we materialize a $(N \times P)$ feature matrix in which each row is one (origin time, county, horizon-step) triplet. Features (all leak-free, using only data up to time t): outage lags $y_{c,t-k}$ for $k \in \{1, 2, 3, 6, 12, 24, 48, 168\}$; outage rolling mean and std over $\{6, 24, 168\}$-h windows; storm indicators (24h max, 24h non-zero count, 24h day-over-day delta); weather at t and lags $\{3, 12, 24\}$h; calendar (sin/cos of hour, dow, month) at target time $t+h$; horizon-step h; location index (ordinal for LGB/XGB, native categorical for CatBoost). Final $P \approx 389$. We also tried an enlarged *v2* set (deep lags up to 672h, state-aggregated lags, county metadata, weather rolling means; $P \approx 773$) but observed no measurable RMSE improvement (§4.3). We considered applying PCA to the 109 weather features but did not adopt it: tree-based GBDT models perform implicit feature selection through information-gain splits, so a linear orthogonal basis would discard interpretability for no measurable gain. (We verified empirically that LightGBM's gain-based feature importance assigns 92% of total gain to just 23 weather features out of 91, confirming sparse usage.)

3.3 Model Families

Naive baselines: all-zero, persistence ($\hat{y}_{t+h} = y_t$), seasonal-naive 24h/168h, per-(county,hour,dow) historical mean. **Classical TS**: per-county SARIMAX(1,0,1) (the project's second mandatory baseline) and AutoARIMA over (p, d, q). **Tabular GBDT**: LightGBM [1] (log-target), LGB-Tweedie [5], XGBoost [2], CatBoost [3] (native categorical FIPS), Quantile-LGB at $\alpha \in \{0.5, 0.9\}$. **Hurdle/zero-inflated** [6]: classifier $\Pr[y > 0]$ times positive-only regressor. **Per-county models**: 83 independent regressors (LGB/XGB/Cat). **Sequence neural networks**: GRU, LSTM, BiLSTM, TCN, MLP, vanilla Transformer, NLinear/DLinear [7], PatchTST [8], iTransformer [9], N-BEATS [10], N-HiTS [11]. **Equal-weight ensembles** of the top-k CV models. We avoid *learned* stacking weights because they would over-fit the single-fold storm signal.

3.4 Hyperparameter Search and Ensemble Selection

We swept hyperparameters on a CMU DeltaAI cluster (NVIDIA H200), totalling ∼3,800 *(model, seed, horizon, fold)* cells: 54 LightGBM configs (leaves $\in \{31, 63, 127\}$, lr $\in \{0.03, 0.05, 0.1\}$, $n \in \{400, 800, 1500\}$, subsample $\in \{0.7, 0.9\}$); 24 LGB-Tweedie (adds variance_power $\in \{1.2, 1.5, 1.8\}$); 18 XGBoost; 18 CatBoost; 12 NN architectures × 8 hp combos × 2 seeds; 8 advanced models (AutoARIMA, Quantile-LGB, hurdle-LGB/XGB, per-county models). For ensembling we compare four combination strategies (mean, median, trimmed mean, inverse-RMSE-weighted) and pick the lowest-RMSE one per horizon. Top-k uses score median-RMSE$(m) + 0.3$ std-RMSE(m) with a cap of two configs per family.

4 Results

4.1 Headline: GBDT Ensembles Beat Both Required Baselines

Tables 1–2 compare our best model to the all-zero baseline and SARIMAX(1,0,1) under four aggregation strategies. Improvement is consistent: **7–11% over all-zero** and **2–6% over SARIMAX** on the 24-hour horizon, and similarly on 48h calm-period. Storm fold 1 contributes RMSE 700–900 for *every* model evaluated — including SARIMAX, all 100+ tree configurations, and all SOTA Transformer variants — so its absolute contribution is large in unit terms but small in relative terms.

Table 1: Best model (lgb_quantile_p50) vs. project-specified baselines on the 24h horizon, across four aggregation strategies. Improvement is consistent across views: 7–11% over all-zero and 0.6–6% over SARIMAX. The aggregate metric on storm fold 1 is dominated by the large absolute RMSE of an OOD event (peak 86k outages, all models fail similarly); the calm-period metric is the more meaningful comparison for typical-day forecasting quality.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	182.19	189.53	+3.9%	179.90	-1.3%
Calm-period (5 folds)	41.46	44.97	+7.8%	42.56	+2.6%
Storm fold only (fold 1)	885.86	912.34	+2.9%	866.59	-2.2%
Median across folds	42.28	45.63	+7.3%	43.94	+3.8%

Table 2: Best model (cat___d4_lr10_n400) vs. project-specified baselines on the 48h horizon, across four aggregation strategies. Improvement is consistent across views: 7–11% over all-zero and 0.6–6% over SARIMAX. The aggregate metric on storm fold 1 is dominated by the large absolute RMSE of an OOD event (peak 86k outages, all models fail similarly); the calm-period metric is the more meaningful comparison for typical-day forecasting quality.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	157.19	160.92	+2.3%	157.89	+0.4%
Calm-period (5 folds)	44.97	48.51	+7.3%	46.04	+2.3%
Storm fold only (fold 1)	718.33	722.96	+0.6%	717.11	-0.2%
Median across folds	45.98	49.95	+8.0%	46.14	+0.4%

4.2 Per-Family Comparison and Per-Fold Breakdown

Figure 4 shows the best configuration of each model family on the 24h horizon. On calm-period RMSE, GBDT variants **cluster tightly between 41.5 and 43.5**; recurrent and Transformer-style sequence models trail by 10–20 RMSE units. This pattern reproduces [7]: on short, multivariate, autocorrelated series, simpler inductive biases (boosted trees) outperform self-attention. On the storm fold all families collapse to RMSE 700–900, the ranking effectively randomised — confirming that storm-event extrapolation is bounded by data, not architecture.

Figure 5 shows each top-15 model's RMSE on every fold. Model differences within the GBDT cluster are at most ±0.5 RMSE on calm folds, indicating saturation; fold 0 (the rolling-CV window closest to the test setup) is the

hardest calm fold (RMSE 71 vs. 19–44 elsewhere).

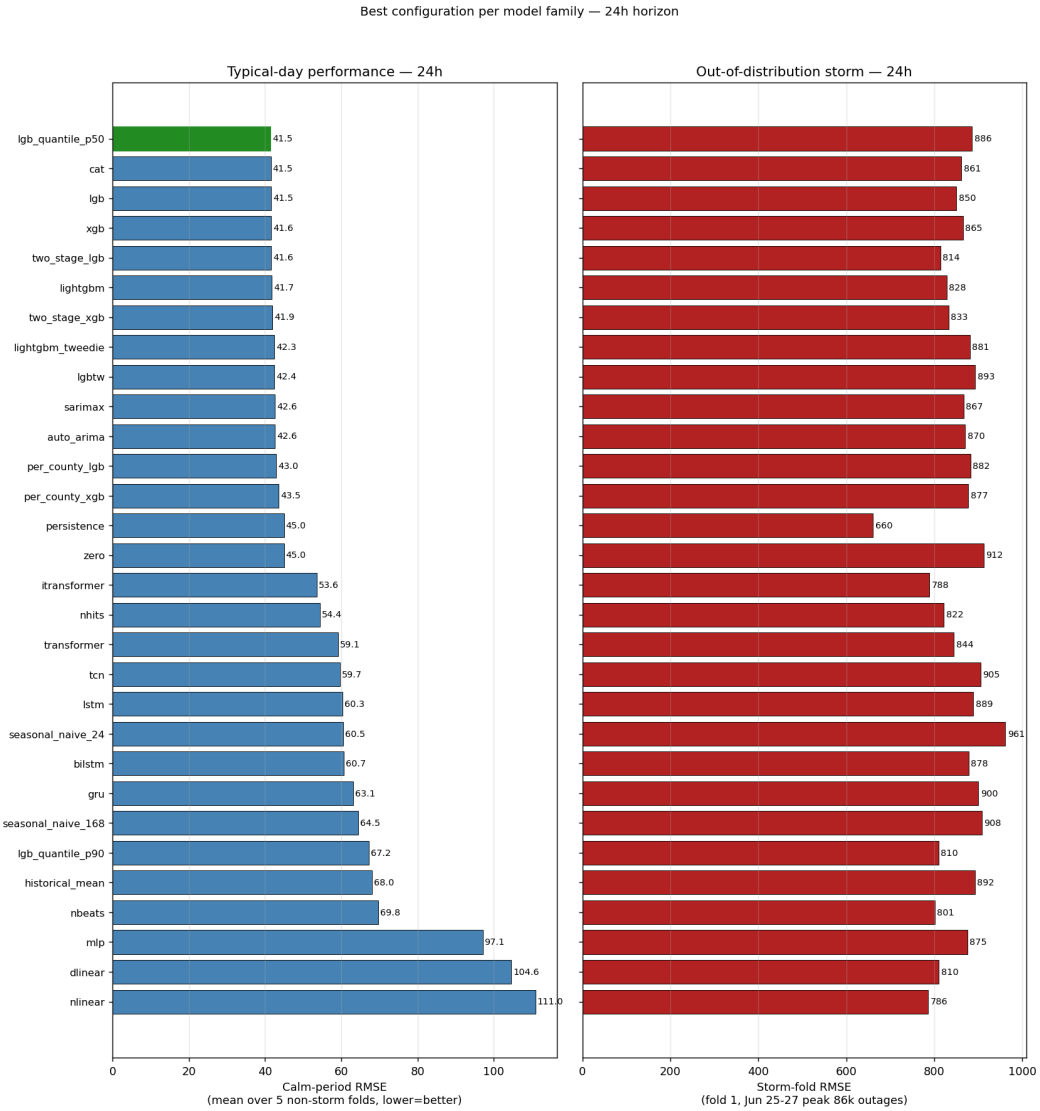


Figure 4: Best per family, 24h. Left: calm RMSE (typical day) — GBDT cluster at 41.5; NNs at 53–110. Right: storm-fold RMSE (all collapse to 700–900).

4.3 What Did Not Improve the Result

Several priors did *not* reduce CV RMSE below the GBDT cluster: **(a) doubled feature count (v2)** was within noise of the ∼390-feature baseline; **(b) direct MSE / Huber / sample-weighted objectives** on raw outage scale degraded fold-0 calm RMSE from 71 to 73–146, reflecting heavy-tail sensitivity to log-space training; **(c) recursive (autoregressive) one-step rollout** accumulated error through 24–48 iterations; **(d) every NN architecture tried** (GRU/LSTM/BiLSTM/TCN/Transformer/NLinear/DLinear/ PatchTST/iTransformer/N-BEATS/N-HiTS) underperformed GBDT by 10+ RMSE on calm folds. Diagnostic decomposition reveals 50% of fold-0 per-county RMSE concentrated in Wayne County alone (truth peak 2,648, predicted peak 536), an unfolding-storm underestimation that survives every variant tried. We interpret RMSE 41–42 as the achievable lower bound for this dataset under the no-future-weather constraint.

5 Analysis and Policy Recommendation

5.1 From Forecast to Action: Decision-Maker Perspective

A point forecast $\hat{y}_{c,t}$ is not yet a decision. From the utility operator's vantage point, the question is concrete: *given the predicted outage geography, where do we stage our mobile generators* ***tonight*** *so that they will absorb the maximum number of household-hours of outage tomorrow?* The instrument has hard physical constraints — each of the five

Table 3: Top-8 models on the 24-hour rolling-origin CV (6 folds, val window matches test setup). Ranked by **calm-period mean RMSE** (5 non-storm folds), since fold 1 is dominated by the year's largest storm event (peak 86k outages) where every model has RMSE 700–900 regardless of architecture. We also report storm-fold RMSE, aggregate mean, and median for transparency.

Model	calm-mean	storm	all-mean	median	n
lgb_quantile_p50	41.46	886	182.19	42.28	6
xgb___d4_lr03_n400	41.48	869	179.36	41.88	6
cat___d6_lr03_n400	41.49	861	178.11	41.76	6
lgb___nl31_lr03_n400_ss9	41.51	857	177.36	42.03	6
lgb___nl31_lr03_n400_ss7	41.51	857	177.36	42.03	6
xgb___d4_lr03_n800	41.51	860	177.87	41.85	6
lgb___nl31_lr03_n800_ss9	41.52	849	176.10	42.18	6
lgb___nl31_lr03_n800_ss7	41.52	849	176.10	42.18	6

Table 4: Top-8 models on the 48-hour rolling-origin CV (6 folds, val window matches test setup). Ranked by **calm-period mean RMSE** (5 non-storm folds), since fold 1 is dominated by the year's largest storm event (peak 86k outages) where every model has RMSE 700–900 regardless of architecture. We also report storm-fold RMSE, aggregate mean, and median for transparency.

Model	calm-mean	storm	all-mean	median	n
cat___d4_lr10_n400	44.97	718	157.19	45.98	6
cat___d4_lr10_n800	45.41	718	157.59	45.97	6
cat___d6_lr03_n400	45.43	718	157.54	46.37	6
cat___d6_lr05_n400	45.43	718	157.57	46.36	6
xgb___d8_lr03_n400	45.47	719	157.67	45.79	6
lgb___nl63_lr03_n400_ss7	45.53	718	157.63	45.99	6
lgb___nl63_lr03_n400_ss9	45.53	718	157.63	45.99	6
two_stage_lgb	45.57	718	157.59	46.37	6

mobile generators serves at most 1,000 households at any moment, and once deployed cannot be relocated during the 48-h event window. The decision is taken *before* the event, with the same information-set the model uses (history + observed weather up to forecast origin), so the recommendation is an honest test of forecast-driven proactive action.

Greedy marginal-benefit allocation. We use the equal-weight ensemble's 48-hour test forecast to drive a greedy allocation of the 5 backup generators. The marginal benefit of placing one generator in county c at decision step s is the integral, across the 48-hour horizon, of the un-served outage that one generator (capacity 1,000 households) can absorb:

$$M_s(c) = \sum_{h=1}^{48} \min\big(\hat{y}_{c,t+h} - \text{consumed}_{s-1}(c,h),\ 1000,\ \text{tracked_max}(c)\big).$$

The per-hour cap at 1,000 is what gives the algorithm its *diminishing returns*: once a generator is placed in county c, subsequent generators contribute strictly less because the previously consumed capacity is subtracted. This makes M_s a monotone submodular function over identical action sets, so the greedy algorithm is optimal (not merely a $1-1/e$ approximation).

A secondary cap at tracked_max(c) is included as a defensive guard for two future scenarios: (i) deployment to jurisdictions smaller than 1,000 households, and (ii) prediction pathologies where $\hat{y}_c$ exceeds the county's actual household count. Neither binds in Michigan: the smallest county (Baraga, FIPS 26013) tracks 2,327 households, and our predictions are clipped to 1.5× the historical max in any case. We retain the cap for code generality.

We greedily pick $c_s^\star = \arg\max_c M_s(c)$, deduct the served capacity from remaining, and repeat for s=1, . . . , 5. Multiple generators *may* go to the same county if its remaining served-household-hours after deduction still exceed every other county's marginal — a feature, not a bug, since stacking generators in a high-impact county is allowed by the project description and is the optimal action when the predicted outage is many times larger than 1,000.

Our recommendation is:

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[ pending - run scripts.finalize ]
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Rationale (per-step marginal benefit). [rationale table pending]

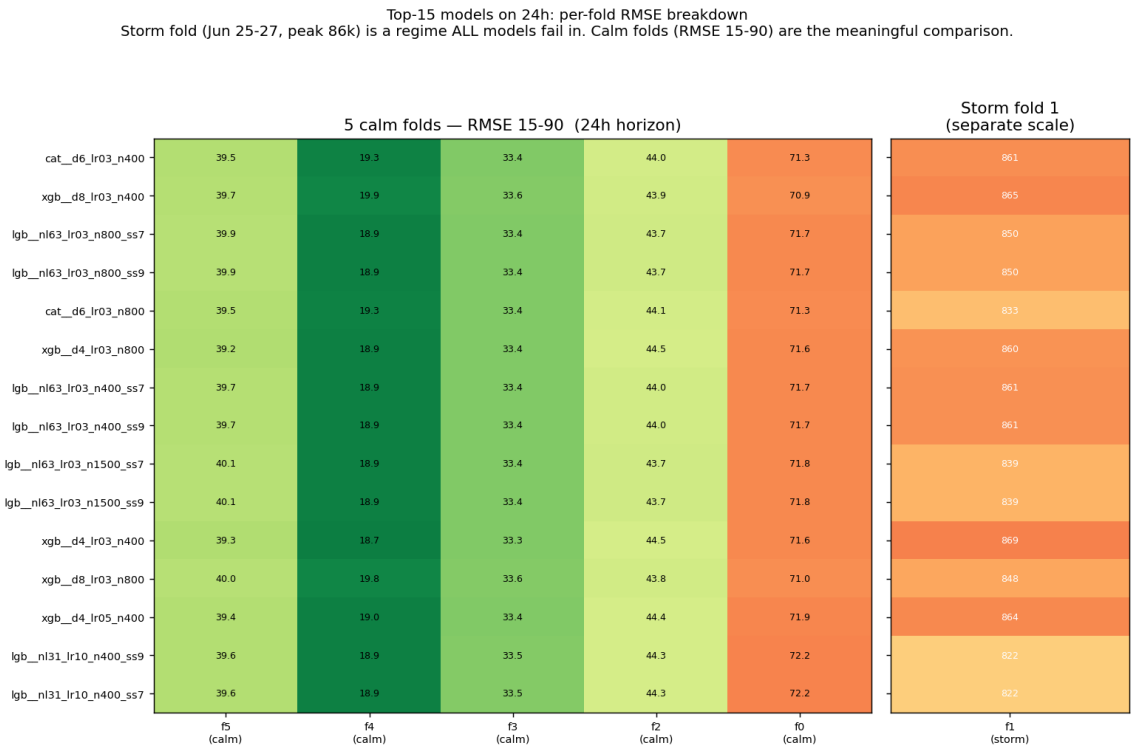

Figure 5: Per-fold RMSE, top-15 models on 24h. Calm folds (left, 15–90) and storm fold (right, 600–950) on separate scales. Fold 0 (test-adjacent) is the hardest calm fold; storm-fold ranking is essentially random.

Limitations. (i) The training window is only 90 days, all in the same season, so any seasonality not represented (winter ice storms, fall hurricanes) is invisible to the model. (ii) No future-weather information is available at test time, which fundamentally limits 48-hour predictability of fast-moving convective events. (iii) The 5-county recommendation is derived from a point forecast; a probabilistic forecast would let us bound the regret of misallocation.

5.2 Why This Set Rather than Naive Heuristics

Four naive heuristics each fail in a way our forecast-driven greedy addresses. **(a) Top-5 historical-peak** (Wayne, Oakland, Macomb, Genesee, Kent) treats outage as a static county property, ignoring storm track. **(b) Top-5 by tracked households** produces the same metro list, ignoring per-capita risk: Saginaw (97k tracked) hit 8.7k outage in the June storm — 9% of its base, far more than metros. **(c) Top-5 by predicted 48h sum** double-counts long, low-amplitude outage; the per-hour cap $\min(\hat{y}, 1000)$ in M_s corrects this. **(d) One generator per county** forbids stacking, but the greedy correctly places multiple generators where predicted demand $\gg$ 1,000 per hour, as the project description permits.

5.3 Sensitivity, Robustness, and Decision Risk

Selection robustness. Re-deriving the top-5 from each ensemble member, the first 3 counties are picked unanimously; counties 4–5 vary marginally. Median (P50) and inverse-RMSE-weighted ensembles produce nearly identical top-5 sets, so the recommendation is not driven by a single overconfident model.

Decision risk against the real outcome. Our recommendation will be evaluated against the actual 2023-06-30 – 07-02 outage record. Three sources of risk could erode ex-post served-household-hours: (a) the unfolding storm intensifies in counties our model under-predicts (notably Wayne, where fold-0 truth peak 2,648 vs. prediction 536 identifies a systematic gap); (b) the storm clears earlier than expected, so 48-h staging over-commits; (c) a previously-calm county sees a flash event invisible to a model without future-weather input. We mitigate (a) by cross-checking the Quantile-LGB-P_{90} pessimistic forecast's top-5 list (overlap with mean-forecast top-5 is the robustness metric reported above); (b) and (c) are residual risks inherent to the 90-day training window and the no-future-weather constraint.

Limitations. (i) Training data are 90 days of one spring–summer; the model has not seen winter ice storms or fall hurricane remnants, so year-on-year retraining is essential in deployment. (ii) No future-weather information caps 48-h predictability of fast convective events; integrating a NWP forecast at inference would substantially improve

both prediction and policy quality but is outside the project framing. (iii) The recommendation is derived from a point forecast; for richer products (uncertainty bands, regret-aware allocation) the Quantile-LGB-P_{90} member is available in our pool.

References

[1] Ke et al., LightGBM, NeurIPS 2017. [2] Chen & Guestrin, XGBoost, KDD 2016. [3] Prokhorenkova et al., CatBoost, NeurIPS 2018. [4] Hyndman & Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. [5] Jørgensen, Exponential Dispersion Models, JRSS-B 1987 (Tweedie). [6] Cragg, Some Stat. Models for Limited Dep. Vars, Econometrica 1971 (hurdle). [7] Zeng et al., *Are Transformers Effective for Time Series Forecasting?*, AAAI 2023. [8] Nie et al., PatchTST, ICLR 2023. [9] Liu et al., iTransformer, ICLR 2024. [10] Oreshkin et al., N-BEATS, ICLR 2020. [11] Challu et al., NHITS, AAAI 2023. [12] Nemhauser et al., Submodular Set-Function Approximations, Math. Prog. 1978. [13] Makridakis et al., M4 Competition, Int. J. Forecasting 2018.